Reproducing and Extending Median Statistics Analysis of the Galactic Rotational Velocity

Edara Yaswanth Balaji
EP22BTECH11007

Arjun Moravekar
EP22BTECH11016

May 1, 2026

Github Repo Link: Click Here

Abstract

We reproduce the median statistics analysis of Camarillo, Dredger & Ratra (2018) for independent measurements of the Milky Way's rotational velocity at the solar radius, Θ_0. The original work compiled independent post-2000 measurements, rescaled them to a common solar radius, separated them by tracer type, and argued that the Old-tracer sample gives the most reliable estimate because it more directly follows the systematic rotation of the Galaxy. In this analysis we implement the same statistical pipeline, verify the main central estimates, reconstruct the normalized-deviation and goodness-of-fit analysis, and extend the same robust-statistics framework to an illustrative comparison set of H_0 measurements. The analysis confirms that Θ_0 is best summarized by a robust median-statistics value near 220 km s^{-1}, with the Old-tracer median giving $\Theta_0 = 219.70$ km s^{-1} and the weighted mean giving 222.01 ± 1.99 km s^{-1}. The report also shows why the median is a useful estimator when measurements are independent but heterogeneous: it is stable against outliers, does not rely on possibly underestimated quoted errors, and remains interpretable when the deviation distribution is mildly non-Gaussian.

1 Introduction

The rotation of the Milky Way is one of the basic ingredients in Galactic astronomy. Two constants are especially important: R_0, the distance from the Sun to the Galactic center, and Θ_0, the rotational velocity of the Milky Way at that radius. These constants enter models of the Galactic potential, the local circular speed, the angular velocity of the Sun, and the calibration of many interstellar and extragalactic measurements. Even a modest change in Θ_0 changes the inferred mass distribution of the Galaxy and the interpretation of the Milky Way rotation curve.

The difficulty is not that Θ_0 has never been measured. The difficulty is that it has been measured many times with different tracers, different assumptions about R_0, different data sets, and different levels of systematic uncertainty. Measurements based on Cepheids, OB stars, open clusters, gas clouds, masers, globular clusters, red clump stars, and large stellar surveys do not all behave as if they are simple draws from one perfectly Gaussian distribution. Some measurements have small quoted uncertainties but are not necessarily free of systematic effects. Some compilations also include repeated or non-independent measurements, which can bias a simple average.

Camarillo, Dredger & Ratra (2018) addressed this problem by compiling independent measurements published since 2000 and applying median statistics. Their central point was that the median is a more robust summary statistic than the weighted mean when the input data may be mildly non-Gaussian or when quoted uncertainties may not capture all sources of error. The weighted

mean is still useful, but it places large weight on measurements with small reported errors. If those errors are underestimated or correlated with other measurements, the weighted mean can become too confident.

This report reproduces and extends the original analysis with three main aims. First, it recalculates the central estimates for Θ_0 using the independent measurements listed by Camarillo et al. (2018), considering the All tracers, Old tracers, and Young tracers groups. Second, it reconstructs the normalized-deviation distributions and compares them with Gaussian, Cauchy, Student's t, and Laplace probability distributions using Kolmogorov-Smirnov tests. Finally, it applies the same robust-statistics approach to a small comparison sample of H_0 measurements, showing how the method performs for another astrophysical quantity with well-known measurement tension.

2 Scientific Background

2.1 Why Θ_0 Matters

The value of Θ_0 describes the circular speed of the Milky Way at the solar radius. Together with R_0, it determines the local angular speed,

$$\omega_0 = \frac{\Theta_0}{R_0}. \tag{1}$$

This angular speed is important because many observations directly constrain angular motion, while the conversion to a linear speed depends on the adopted distance scale. A consistent estimate of Θ_0 therefore requires careful treatment of R_0.

The original paper adopted the recent median-statistics value $R_0 = 7.96 \pm 0.17$ kpc from a related analysis. The practical summary from Camarillo et al. (2018) was

$$\Theta_0 = 220 \pm 7 \text{ km s}^{-1} \tag{2}$$

at 1σ, and

$$\Theta_0 = 220 \pm 10 \text{ km s}^{-1} \tag{3}$$

at 2σ, with

$$\omega_0 = 27.6 \pm 1.1 \text{ km s}^{-1} \text{ kpc}^{-1}. \tag{4}$$

The goal of the present analysis was to reproduce the statistical reasoning behind this result.

2.2 Why Median Statistics Is Appropriate

If every measurement were independent, Gaussian, and accompanied by a perfectly estimated uncertainty, the weighted mean would be the natural estimator. In practice, astronomical measurements often include hidden systematic effects. The median is useful in this setting because it depends mainly on the ordering of the data and not on the exact magnitude of extreme values. Its breakdown point is 50%, meaning that nearly half of the measurements would have to be corrupted before the median itself becomes unreliable.

Median statistics also separates two ideas that are sometimes mixed together:

- the best central value implied by the rank ordering of independent measurements;
- the shape of the error or deviation distribution around that central value.

This separation is useful for the Θ_0 problem. The central estimate can remain stable even if the deviations are mildly non-Gaussian, and the deviation distribution can then be tested separately.

3 Data

3.1 Θ_0 Measurements

The analysis uses the 29 independent post-2000 measurements listed in Camarillo et al. (2018). These measurements were taken from the paper's Table 1 after rescaling them to a common value of R_0. The measurements are separated into three analysis groups:

- **All tracers**: all 29 independent measurements.
- **Old tracers**: 18 Old-tracer measurements plus the McMillan (2017) measurement classified as Both, giving $N = 19$.
- **Young tracers**: 10 Young-tracer measurements plus the McMillan (2017) measurement classified as Both, giving $N = 11$.

The Old-tracer group is the physically preferred group in the original paper. Old tracers are expected to better follow the systematic rotation of the Galaxy. Young tracers, such as Cepheids and OB stars, may be affected by peculiar motions, local streaming, or other non-circular motions. Young-tracer measurements remain informative, but they should not automatically be treated as equally representative of the underlying circular speed.

The rescaling formula used by the original paper can be written as

$$\Theta_{0,i}^{\mathrm{res}} = \frac{R_0 \Theta_{0,i}}{R_{0,i}}, \tag{5}$$

with uncertainty propagated as

$$\sigma_{\Theta_{0,i}^{\mathrm{res}}} = \Theta_{0,i}^{\mathrm{res}} \sqrt{\left(\frac{\sigma_{R_0}}{R_0}\right)^2 + \left(\frac{\sigma_{\Theta_{0,i}}}{\Theta_{0,i}}\right)^2 + \left(\frac{\sigma_{R_{0,i}}}{R_{0,i}}\right)^2}. \tag{6}$$

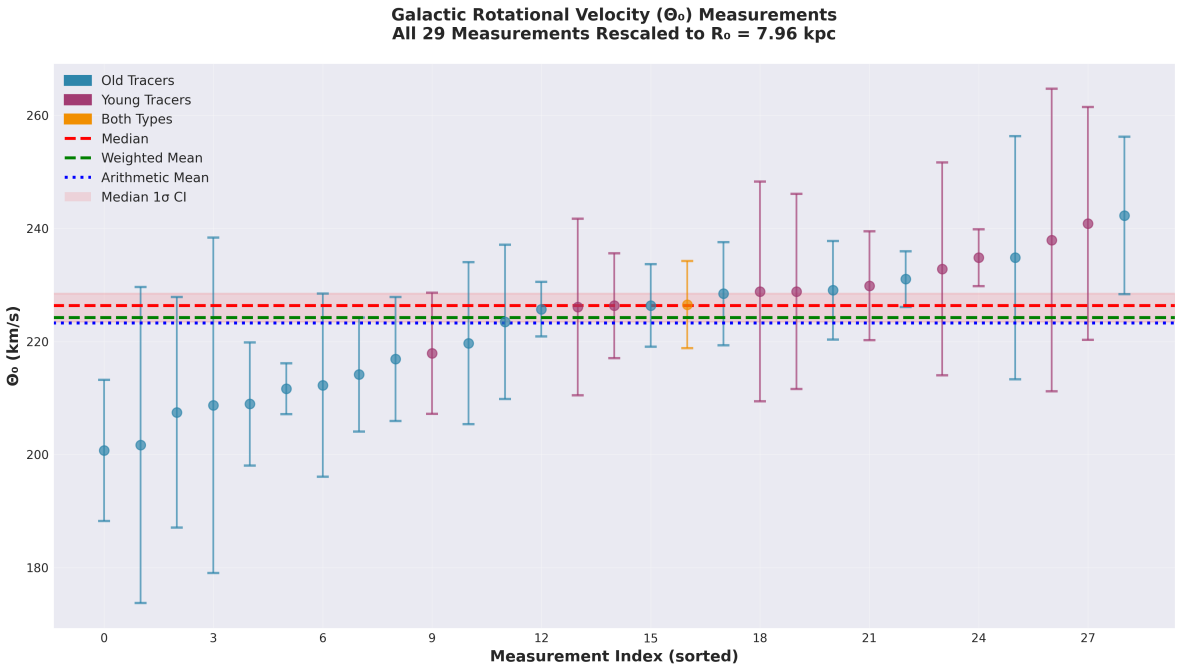


Figure 1: Overview of the 29 independent rescaled Θ_0 measurements. The plot shows both the spread of the measurements and the role of the quoted uncertainties. The measurements cluster around 220–230 km s^{-1}, but the scatter is large enough that robust statistics are useful.

3.2 H_0 Extension Data

As an extension beyond the original paper, the analysis also includes 14 H_0 measurements. This is not intended to replace a full cosmological meta-analysis. Instead, it provides a useful comparison problem because H_0 measurements have a well-known method-dependent tension. Early-universe measurements such as Planck and late-universe distance-ladder measurements do not sit comfortably on one narrow Gaussian distribution. The H_0 extension therefore tests whether the same workflow is portable:

- compute robust and weighted central estimates;
- examine normalized deviations;
- compare Gaussian and heavy-tailed models;
- interpret whether the median gives a more stable summary than the weighted mean.

4 Methodology

4.1 Central Estimates

For each data subset, the analysis computes three central estimates: the arithmetic mean, the weighted mean, and the median. The arithmetic mean is useful as a simple reference value,

$$\bar{\Theta}_0 = \frac{1}{N}\sum_{i=1}^{N}\Theta_{0,i}. \tag{7}$$

The weighted mean is

$$\Theta_0^{\rm wm} = \frac{\sum_{i=1}^N \Theta_{0,i}/\sigma_i^2}{\sum_{i=1}^N 1/\sigma_i^2}, \tag{8}$$

with formal uncertainty

$$\sigma_{\Theta_0}^{\rm wm} = \frac{1}{\sqrt{\sum_{i=1}^N 1/\sigma_i^2}}. \tag{9}$$

The median is the central ranked value of the measurements. Unlike the weighted mean, it does not use the quoted measurement uncertainties. This is a feature, not only a limitation, when the reliability of those uncertainties is part of the problem.

4.2 Median Confidence Intervals

The uncertainty on the median was computed using the binomial median-statistics method of Gott et al. (2001). For N independent measurements sorted by value, the probability associated with a given median rank is given by a binomial expression. If i is the number of measurements below the true median, this probability is

$$P_i = 2^{-N} \frac{N!}{i!(N-i)!}. \tag{10}$$

The 1σ median interval is then selected so that it encloses approximately 68.27% of the total probability, and the 2σ interval encloses approximately 95.45%. Since the data are discrete, the interval endpoints are actual ranked measurements, and the enclosed probability can only approximate the target value.

4.3 Normalized Deviations

After choosing a central estimate, the next step is to construct an error distribution. For a general central estimate $\Theta_0^{\rm CE}$, the normalized deviation of measurement i is

$$N_{\sigma,i} = \frac{\Theta_{0,i} - \Theta_0^{\rm CE}}{\sqrt{{\rm Var}(\Theta_{0,i} - \Theta_0^{\rm CE})}}. \tag{11}$$

For the median central estimate, the analysis uses the same form as the original paper:

$$N_{\sigma,i}^{\rm med} = \frac{\Theta_{0,i} - \Theta_0^{\rm med}}{\sqrt{\sigma_i^2 + (\sigma_{\Theta_0}^{\rm med})^2}}. \tag{12}$$

The plus sign appears because the median estimate is treated as independent of the individual measurement uncertainty in this approximation.

For the weighted mean central estimate, the central estimate is correlated with the data used to compute it. The original paper therefore uses

$$N_{\sigma,i}^{\rm wm} = \frac{\Theta_{0,i} - \Theta_0^{\rm wm}}{\sqrt{\sigma_i^2 - (\sigma_{\Theta_0}^{\rm wm})^2}}. \tag{13}$$

The minus sign is important. It reflects the fact that the weighted mean is not independent of the measurements it summarizes.

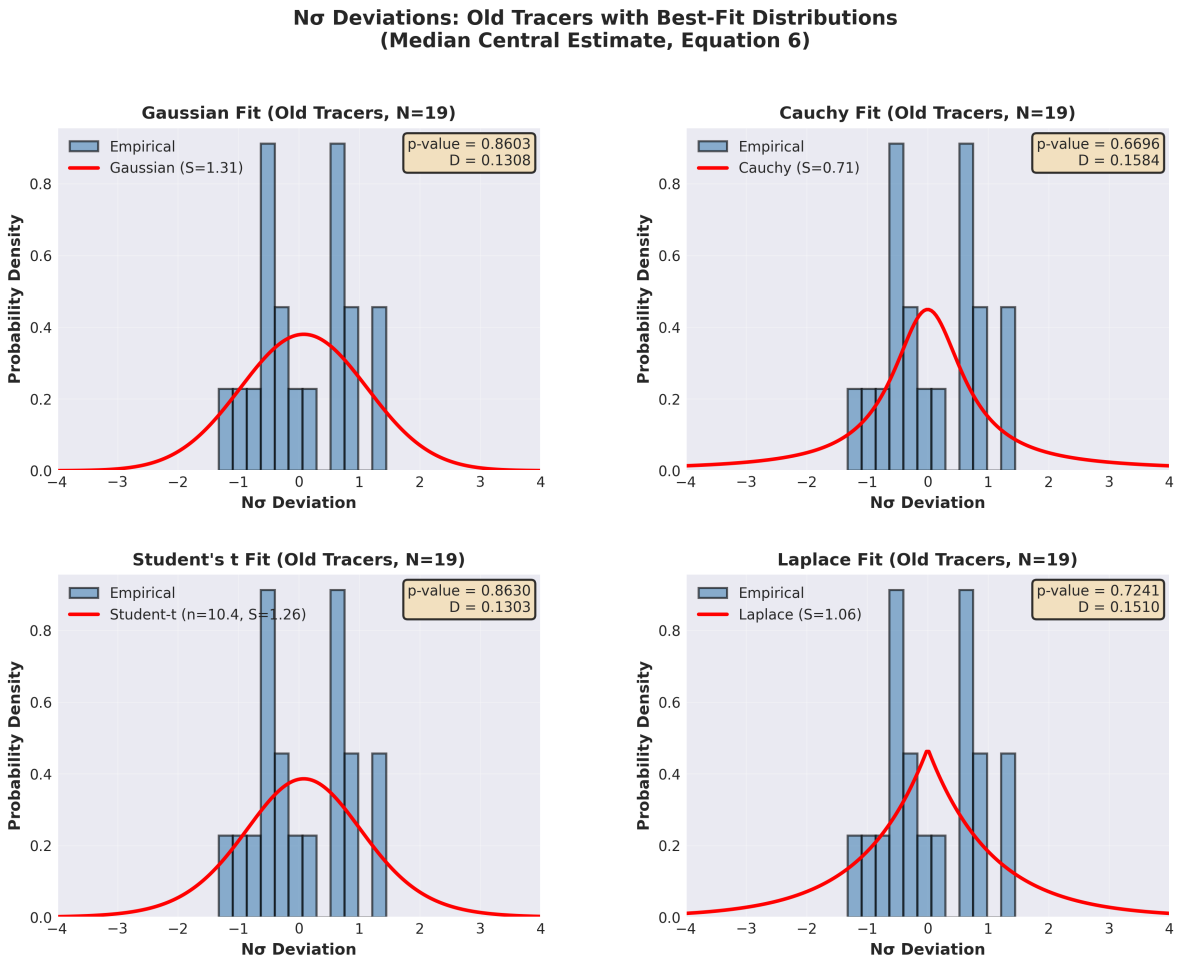


Figure 2: Normalized deviations for the Θ_0 measurements. This diagnostic plot is the bridge between the central-estimate calculation and the goodness-of-fit analysis. A perfectly Gaussian measurement set would produce a deviation distribution close to the standard normal form.

4.4 Distribution Fits

The normalized deviations were compared with four signed probability density functions. The standard Gaussian model is

$$f(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right). \tag{14}$$

The Cauchy model is

$$f(x) = \frac{1}{\pi(1+x^2)}. \tag{15}$$

The Student's t model is

$$f(x) = \frac{\Gamma[(n+1)/2]}{\sqrt{\pi n}\,\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}. \tag{16}$$

The Laplace model is

$$f(x) = \frac{1}{2} \exp(-|x|). \tag{17}$$

These four models were chosen because they represent different tail behaviors. The Gaussian is the usual benchmark. The Cauchy and Student's t distributions have heavier tails. The Laplace

distribution is more sharply peaked at the center while still having heavier tails than a Gaussian. In the original paper, a scale factor S is varied to find the best agreement with the empirical distribution.

4.5 Kolmogorov-Smirnov Test

The Kolmogorov-Smirnov test compares the empirical cumulative distribution function $F_{\text{emp}}(x)$ with a theoretical cumulative distribution function $F_{\text{theo}}(x)$:

$$D = \max_x |F_{\text{emp}}(x) - F_{\text{theo}}(x)|. \tag{18}$$

The D statistic measures the largest vertical distance between the two CDFs. The p-value is then computed from the Kolmogorov distribution. A large p-value means the tested distribution cannot be rejected as a plausible description of the data. It does not prove that the tested distribution is the true data-generating model, especially here because the distribution parameters are fitted from the same small samples being tested.

In this analysis the KS machinery uses a standard two-sided empirical-CDF statistic. The code uses `scipy` for the Student's t distribution and for KS survival probabilities, while keeping the data handling, central-estimate calculations, fitting loop, bootstrap analysis, and plotting code explicit in the project files and notebook.

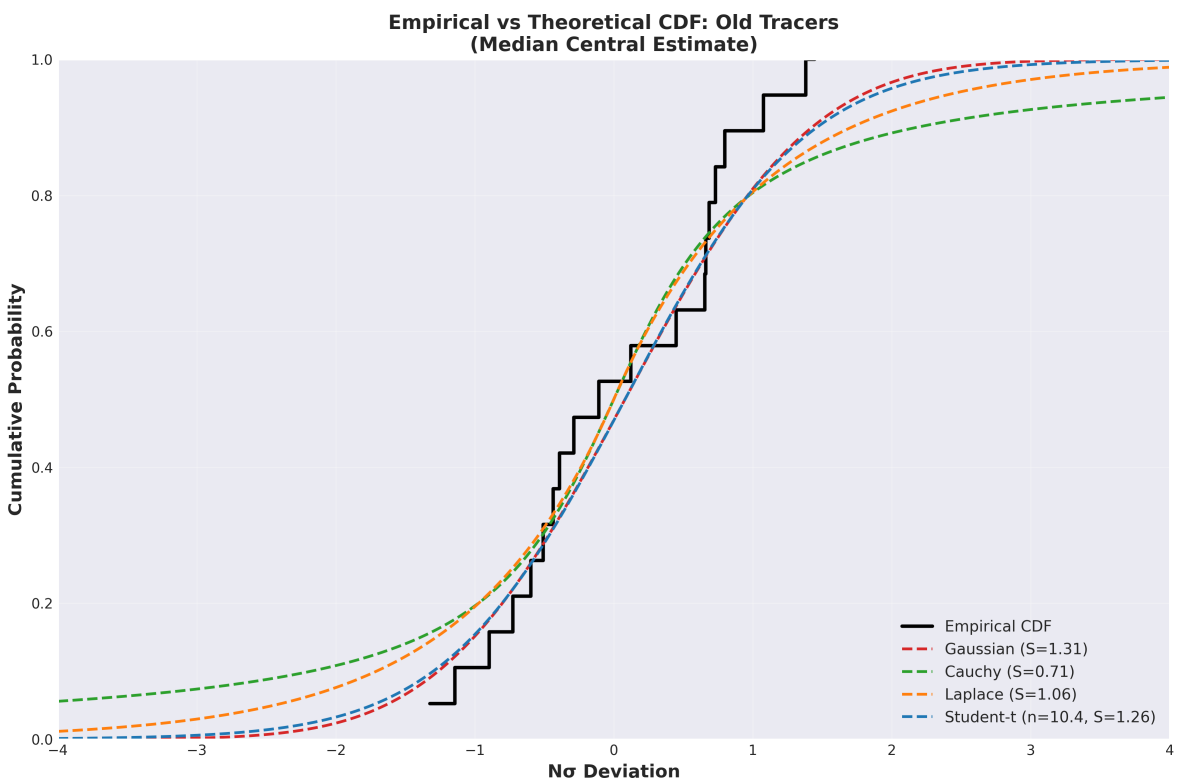


Figure 3: Empirical CDFs compared with fitted theoretical CDFs for the Θ_0 normalized deviations. The CDF view is useful because the KS statistic is defined by the maximum separation between the empirical and theoretical CDFs.

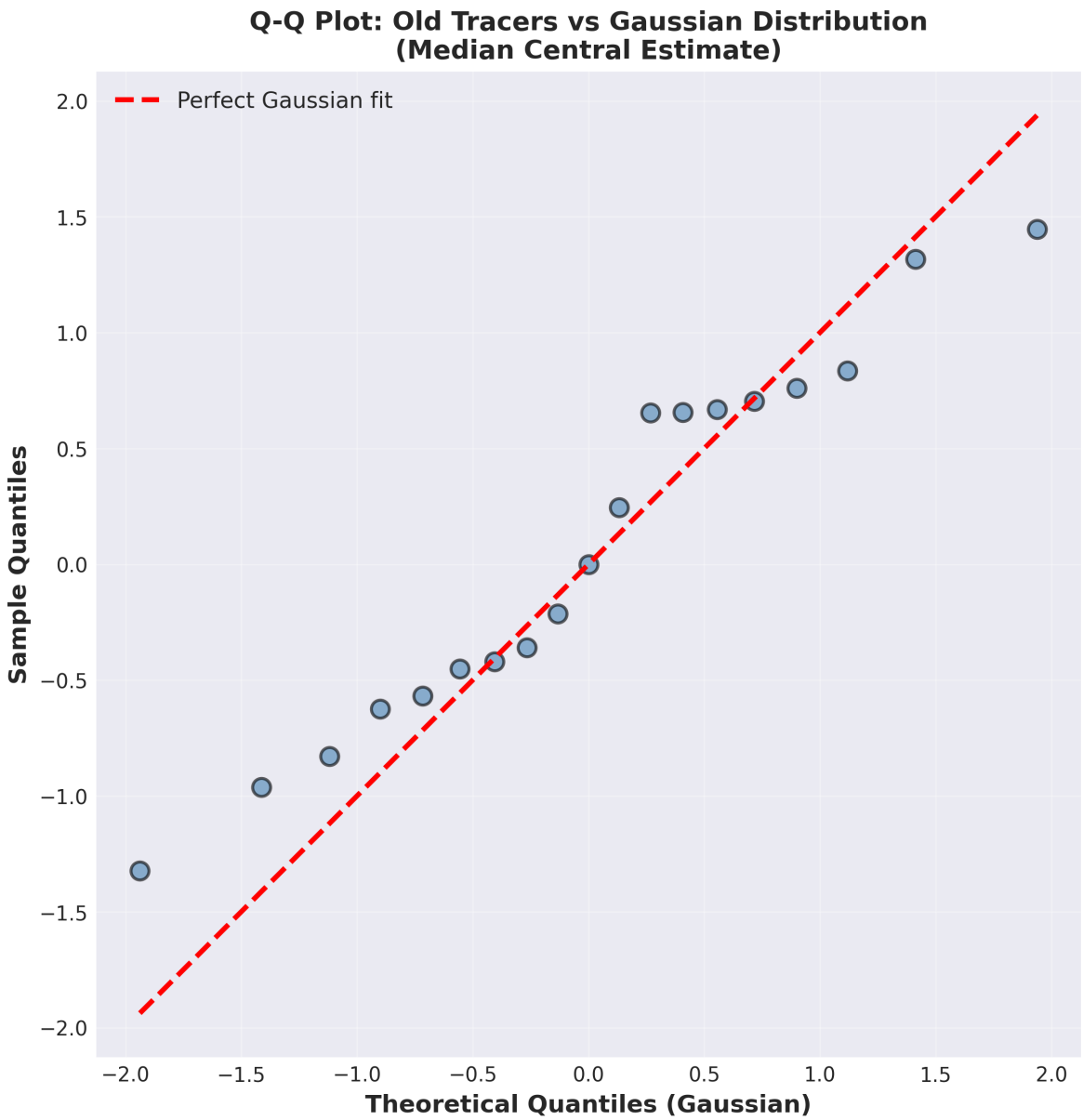


Figure 4: Q-Q diagnostic plot for the Θ_0 deviation distribution. Points close to the reference line indicate agreement between empirical and theoretical quantiles. Departures in the outer quantiles reveal tail behavior.

4.6 Bootstrap Analysis

The bootstrap analysis estimates how stable the central estimates are under resampling. Each bootstrap sample is created by drawing N measurements with replacement from the original sample. The statistic of interest, such as the median or weighted mean, is then recomputed. Repeating this process 10,000 times gives a sampling distribution for the estimator.

The bootstrap is useful here because the Θ_0 sample is small and heterogeneous. It does not assume that the original data are exactly Gaussian. Instead, it asks how much the answer changes when the available measurements are resampled.

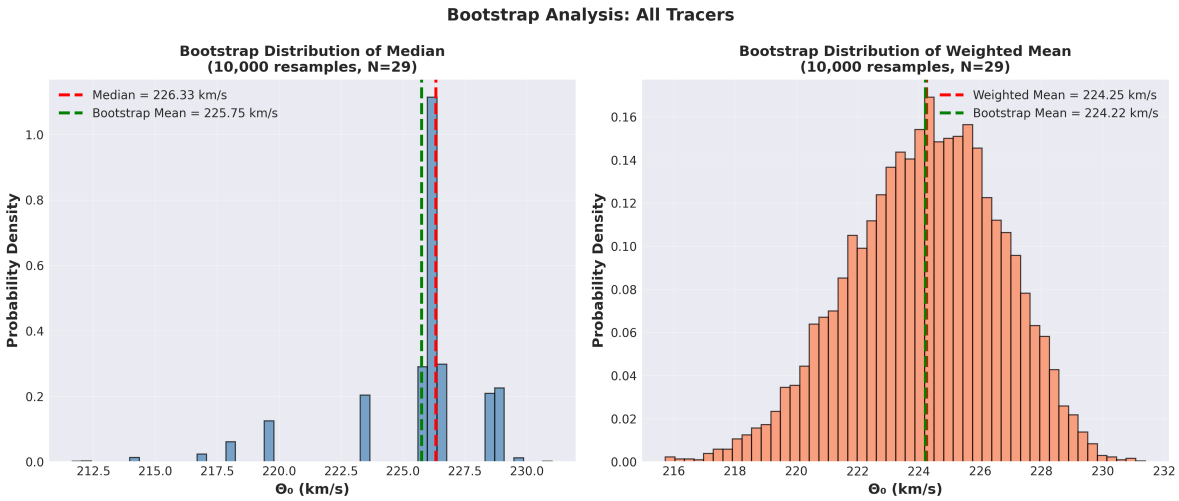


Figure 5: Bootstrap distributions for the Θ_0 central estimates. The width of each bootstrap distribution gives an empirical measure of estimator stability under resampling.

4.7 Implementation Approach

The analysis was implemented in four main Python modules:

- `data.py`: stores the rescaled Θ_0 data, tracer labels, references, and the H_0 extension data.
- `stats_utils.py`: implements probability distributions, CDFs, special-function approximations, KS tests, chi-squared tests, Anderson-Darling statistics, bootstrap resampling, and model-selection utilities.
- `analysis.py`: computes medians, weighted means, confidence intervals, normalized deviations, fitted distributions, and bootstrap summaries.
- `plotting.py`: creates the diagnostic figures used in this report.

The required Python dependencies are `numpy`, `pandas`, `matplotlib`, `seaborn`, and `scipy`. The companion notebook installs these libraries at the beginning so the analysis can be rerun in Google Colab.

5 Results for Θ_0

5.1 Central Estimates

Table 1 summarizes the reproduced central estimates. The most important result is the Old-tracer median of 219.70 km s^{-1}, because this is the subset emphasized by Camarillo et al. (2018). The All-tracer median is higher, 226.33 km s^{-1} with the rounded table values used here, because the All-tracer sample includes Young tracers that tend to give larger values. The Young-tracer median is 228.85 km s^{-1}.

Table 1: Central estimates for rescaled Θ_0 measurements. Values are in km s^{-1}. The asymmetric median errors come from rank-based median statistics, while the weighted-mean errors come from Eq. 9.

Sample	N	Arithmetic Mean	Median	Weighted Mean
All tracers	29	223.25 ± 2.10	$226.33^{+2.13}_{-2.87}$	224.25 ± 1.71
Old tracers	19	219.47 ± 2.65	$219.70^{+6.66}_{-7.43}$	222.01 ± 1.99
Young tracers	11	230.08 ± 1.91	$228.85^{+3.98}_{-2.33}$	230.04 ± 3.10

The weighted means match the paper values very closely:

- All tracers: this analysis gives 224.25 ± 1.71 km s^{-1}; the paper gives 224.36 ± 1.67 km s^{-1}.
- Old tracers: this analysis gives 222.01 ± 1.99 km s^{-1}; the paper gives 222.01 ± 1.99 km s^{-1}.
- Young tracers: this analysis gives 230.04 ± 3.10 km s^{-1}; the paper gives 230.05 ± 3.09 km s^{-1}.

The median values also reproduce the paper's central values to rounding accuracy. The preferred Old-tracer median remains near 220 km s^{-1}, which is the main physical conclusion of the original analysis.

5.2 Old and Young Tracers

The Old and Young tracer subsets behave differently. The Old-tracer median is 219.70 km s^{-1}, while the Young-tracer median is 228.85 km s^{-1}. This separation is important because it supports the original paper's decision to favor the Old-tracer result. Young tracers are more likely to include peculiar motions that do not perfectly represent circular Galactic rotation.

The weighted means tell a similar story. The Old-tracer weighted mean is 222.01 km s^{-1}, while the Young-tracer weighted mean is 230.04 km s^{-1}. The fact that both median and weighted-mean summaries show a higher Young-tracer value makes the subset difference more meaningful than a single outlier effect.

5.3 Goodness-of-Fit Results

For the Old-tracer deviation distribution using the median central estimate, the analysis obtains the KS results shown in Table 2. The Gaussian and Student's t fits give the largest descriptive p-values, near 0.86, while the Cauchy and Laplace fits also remain acceptable by the KS criterion. The scientific conclusion is the same as in the original paper: the Old-tracer sample is only mildly non-Gaussian, and no tested model is strongly rejected by these small-sample diagnostics.

Table 2: KS goodness-of-fit results for Old tracers using the median central estimate.

Distribution	Best parameter	D statistic	p-value
Gaussian	$S = 1.314$	0.1308	0.8603
Cauchy	$S = 0.708$	0.1584	0.6696
Student's t	$n = 10.4,\ S = 1.263$	0.1303	0.8630
Laplace	$S = 1.061$	0.1510	0.7241

The original paper reported, for the Old-tracer median-deviation case, $p = 0.83$ for a Gaussian with $S = 1.08$, $p = 0.71$ for a Cauchy with $S = 0.70$, $p = 0.83$ for Student's t with $n = 36$ and

$S = 1.07$, and $p = 0.76$ for a Laplace distribution with $S = 1.04$. The reproduced numerical details are not identical because this report uses the rounded table values, a standard two-sided KS statistic, and fitted-parameter p-values that should be read descriptively. The overall interpretation is unchanged: the Old-tracer sample is at most mildly non-Gaussian, and no strong evidence appears for a large population of undiscovered systematic errors.

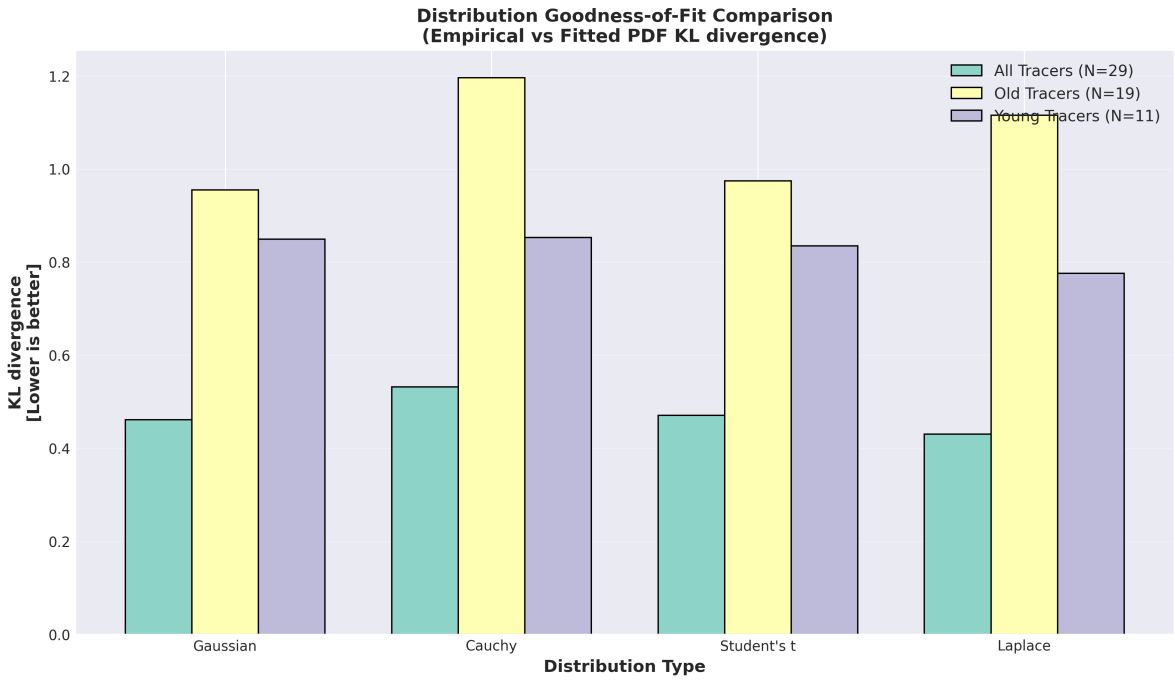


Figure 6: Additional distribution-comparison diagnostic for the Θ_0 samples. The analysis uses this as a supporting metric alongside KS tests rather than as the primary result.

The supporting diagnostic in Figure 6 is based on the Kullback-Leibler divergence, which measures how much information is lost when a fitted model distribution Q is used to approximate the empirical distribution P:

$$D_{\mathrm{KL}}(P\|Q) = \sum_i P_i \log\left(\frac{P_i}{Q_i}\right). \tag{19}$$

Lower values indicate closer agreement between the empirical histogram and the fitted distribution. This metric is used only as a supporting diagnostic because it depends on histogram binning and is less directly tied to the original paper than the KS tests.

5.4 Bootstrap Interpretation

The bootstrap resampling confirms that the median estimate is stable but has a broader sampling distribution than the formal weighted mean. The median does not use uncertainty weights, so it gives up some formal precision in exchange for robustness. The weighted mean gives a smaller formal uncertainty, but that precision depends strongly on the quoted measurement errors being reliable.

For this analysis, the bootstrap standard deviations for the median and weighted mean are 2.66 and 2.44 km s^{-1} for All tracers, 5.05 and 2.79 km s^{-1} for Old tracers, and 2.20 and 2.26 km s^{-1} for Young tracers. These values are of the same general scale as the analytic uncertainties, which

is a useful validation check. They show that the central estimates are not being driven by one pathological resampling behavior.

6 Extension to H_0 Measurements

6.1 Motivation

The H_0 extension was included because the Hubble constant is another example where combining measurements is not a purely mechanical averaging problem. Measurements from the early universe and late universe can give visibly different results. This makes H_0 a useful comparison case for median statistics, even though the physical context is different from Galactic rotation.

The extension data include 14 measurements spanning roughly 66.93 to 74.30 km s^{-1} Mpc^{-1}. The arithmetic mean is 70.46 km s^{-1} Mpc^{-1} and the median is 70.20 km s^{-1} Mpc^{-1}. The formal weighted mean is 68.30 ± 0.28 km s^{-1} Mpc^{-1} for the stored data, because the low-H_0 CMB measurements have comparatively small quoted uncertainties and therefore receive large weights.

This is a case where reporting both the median and the weighted mean is informative. The median gives the central rank of the measurement collection; the weighted mean shows what happens when quoted uncertainties are accepted at face value. The result should not be interpreted as a consensus estimate of H_0, because the extension data are illustrative and do not model correlations, shared calibrations, or method-specific covariance.

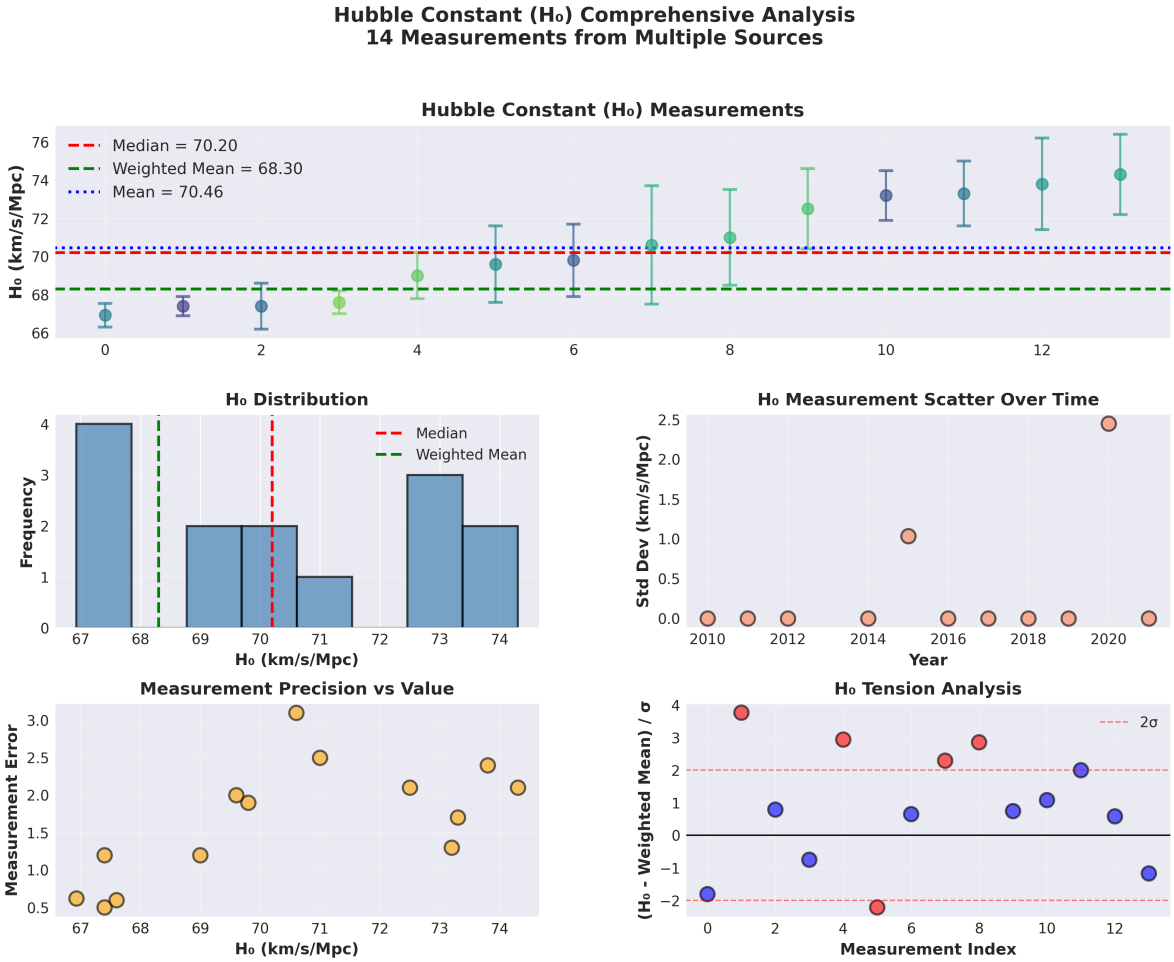
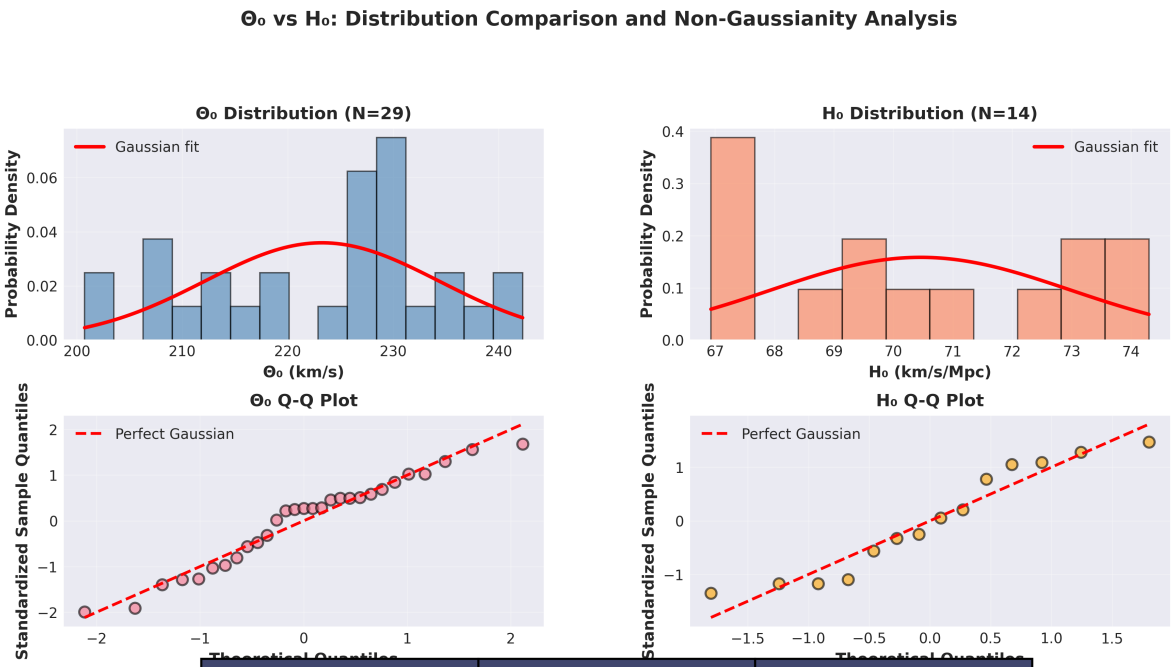


Figure 7: Comprehensive H_0 extension analysis. The panels summarize the measurement distribution, central estimates, deviations, and tension structure in the comparison data set. The H_0 case highlights how strongly a weighted mean can be pulled by small quoted uncertainties.

6.2 Comparison with Θ_0

The Θ_0 and H_0 examples share the same statistical lesson: a set of published measurements may be independent and useful without being well described by one simple Gaussian distribution. In both cases, the median gives a stable central summary, while the weighted mean tests the implications of trusting quoted error bars.

There is also an important difference. The Θ_0 Old-tracer data are only mildly non-Gaussian, and the median and weighted mean are reasonably close. For H_0, the weighted mean is pulled toward the low end of the sample because the most precise measurements are also systematically low compared with several distance-ladder measurements. This makes the H_0 example a stronger demonstration of why robust methods should be reported alongside weighted means.



Statistic	Θ₀ (All 29)	H₀ (All 14)
Mean	223.25	70.46
Median	226.33	70.20
Std Dev	11.10	2.52
Min	200.74	66.93
Max	242.28	74.30
Range	41.54	7.37

Figure 8: Comparison between the Θ_0 and H_0 analyses. The figure shows that the same statistical workflow can be used across different astrophysical measurement compilations, while the interpretation depends on the physical origin of each data set.

7 Comparison with the Original Paper

The reproduction recovers the central values that define the original paper's conclusion. The Old-tracer median is reproduced as 219.70 km s^{-1}, and the Old-tracer weighted mean is reproduced as 222.01 ± 1.99 km s^{-1}, matching the published value. The All-tracer and Young-tracer weighted means also agree to within rounding-level differences.

The strongest scientific conclusion of Camarillo et al. (2018) is therefore preserved. The independent Old-tracer measurements are only mildly non-Gaussian and support the practical estimate

$$\Theta_0 \simeq 220 \pm 7 \ \mathrm{km\,s^{-1}} \tag{20}$$

at 1σ, or

$$\Theta_0 \simeq 220 \pm 10 \ \mathrm{km\,s^{-1}} \tag{21}$$

at 2σ.

The difference between this estimate and some earlier compilations is mainly methodological. Compilations that include non-independent measurements, repeated tracer classes, or mixed

Old/Young samples can shift the result toward higher values near 230 km s^{-1}. The present reproduction follows the original paper's emphasis on independence and tracer classification.

Table 2: Central Estimates (Median Statistics)

Tracers	N	Median (1σ CI)	Median (2σ CI)	Weighted Mean
All	29	226.33 +2.13/-2.87	+2.52/-8.42	224.25 ± 1.71
Old	19	219.70 +6.66/-7.43	+8.76/-10.75	222.01 ± 1.99
Young	11	228.85 +3.98/-2.33	+9.09/-2.76	230.04 ± 3.10

Table 3: Distribution Fits - Old Tracers (N=19, Median Central Estimate)

Distribution	Scale Factor S	KS p-value	KS D-statistic
Gaussian	1.314	0.8603	0.1308
Cauchy	0.708	0.6696	0.1584
Student's t	1.263 (n=10.4)	0.8630	0.1303
Laplace	1.061	0.7241	0.1510

Figure 9: Generated summary table of the main project results. This figure is useful as a compact visual reference for the central estimates and goodness-of-fit outputs.

8 Discussion

8.1 Robustness of the Median

The median is not automatically better than the weighted mean in every situation. It is better suited to this problem because the reliability of individual error bars is part of the uncertainty. The weighted mean answers the question: what central value is preferred if the quoted uncertainties are trusted as relative weights? The median answers a different question: what central value is preferred by the ordering of independent measurements?

For Θ_0, these two answers are close enough that the result is reassuring. The Old-tracer median is 219.70 km s^{-1} and the Old-tracer weighted mean is 222.01 km s^{-1}. This agreement means the preferred estimate is not an artifact of one method. At the same time, the median remains the safer quoted summary because it is less sensitive to any single underestimated error bar.

8.2 Meaning of Mild Non-Gaussianity

The deviation distributions do not show strong disagreement with Gaussianity. Instead, the evidence points to mild non-Gaussianity. This is important. If the data were strongly non-Gaussian, one

might suspect large hidden systematics or an invalid combination of measurements. If the data were perfectly Gaussian, the weighted mean might be fully adequate. The actual result lies between these extremes: the Old-tracer sample is close enough to Gaussian that current error bars are broadly reasonable, but not so perfectly Gaussian that robust methods should be ignored.

This is why the original paper's conclusion is measured rather than dramatic. The result does not claim that previous measurements are wrong. It argues that the most reliable summary of the independent Old-tracer collection is a median-statistics estimate near 220 km s^{-1}.

8.3 Old Tracers as the Preferred Sample

The Old-tracer preference is both statistical and physical. Statistically, the Old-tracer subset produces a lower and more stable value than the Young-tracer subset. Physically, Old tracers are less likely to be dominated by short-timescale peculiar motions. Young tracers can be valuable probes, but their kinematics may include local effects that are not the same as the long-term circular speed of the Galaxy.

The McMillan (2017) measurement is classified as Both and is included in both Old and Young subsets. This follows the original paper and keeps the subset definitions consistent with the source analysis.

8.4 Lessons from the H_0 Extension

The H_0 extension shows that the same method can be applied outside the Galactic-rotation problem, but it also shows why interpretation matters. For Θ_0, the median and weighted mean are fairly close. For H_0, the median and weighted mean differ more strongly because the low-H_0 measurements have small quoted uncertainties. The robust median therefore provides a complementary summary of the measurement collection, while the weighted mean exposes the dominance of a subset of precise measurements.

This does not resolve the Hubble tension. It simply demonstrates that robust statistics can clarify the structure of the disagreement.

9 Limitations

This work is a reproduction and extension of a published statistical analysis, not a new observational compilation. The Θ_0 data are taken from the original paper's rescaled values, and the conclusions depend on the independence classification used there. If future work revises the independence of some measurements or adds new measurements, the analysis should be rerun.

The distribution-fitting results should also be interpreted with care. The sample sizes are small: $N = 29$ for All tracers, $N = 19$ for Old tracers, and $N = 11$ for Young tracers. With small samples, KS tests have limited power, and several distributions can appear acceptable. This is why the report emphasizes robust central estimates and qualitative deviation behavior rather than over-interpreting one best-fit distribution.

Finally, the H_0 extension is illustrative. A complete H_0 meta-analysis would need a more careful treatment of correlated measurements, method classes, covariance, and shared calibration data.

10 Conclusion

This report reproduces and extends the median-statistics analysis of Camarillo, Dredger & Ratra (2018). The main results are:

1. The preferred Old-tracer median is $\Theta_0 = 219.70$ km s^{-1}, supporting the practical summary $\Theta_0 \simeq 220$ km s^{-1}.

2. The Old-tracer weighted mean is 222.01 ± 1.99 km s^{-1}, matching the published value to the displayed precision.

3. The All-tracer and Young-tracer samples give higher central values, showing the importance of tracer classification.

4. The normalized-deviation distributions are only mildly non-Gaussian, so robust statistics are useful but the data do not demand large undiscovered systematic errors.

5. Bootstrap resampling supports the stability of the central estimates.

6. The H_0 extension demonstrates that the same workflow is useful for other heterogeneous astrophysical measurement collections.

The central lesson is that robust statistics do not replace physical judgment. They work best when combined with careful data selection, independence checks, and an understanding of the tracer populations. For the Milky Way rotational velocity, that combination leads to a reliable and interpretable estimate of $\Theta_0 \approx 220$ km s^{-1}.

A Reproducibility and Dependencies

The companion notebook `Theta0_H0_Reproducible_Analysis.ipynb` contains the complete reproducible workflow for this report. It includes the Θ_0 and H_0 data tables inline, installs the required Google Colab dependencies, runs the central-estimate and distribution-fitting calculations, and regenerates all ten figures used in the report.

The required Python packages are `numpy`, `pandas`, `matplotlib`, `seaborn`, and `scipy`. The local project code is organized into `data.py`, `stats_utils.py`, `analysis.py`, `plotting.py`, and `main.py`. Running `main.py` regenerates the same plots and numerical summaries used in this paper.

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